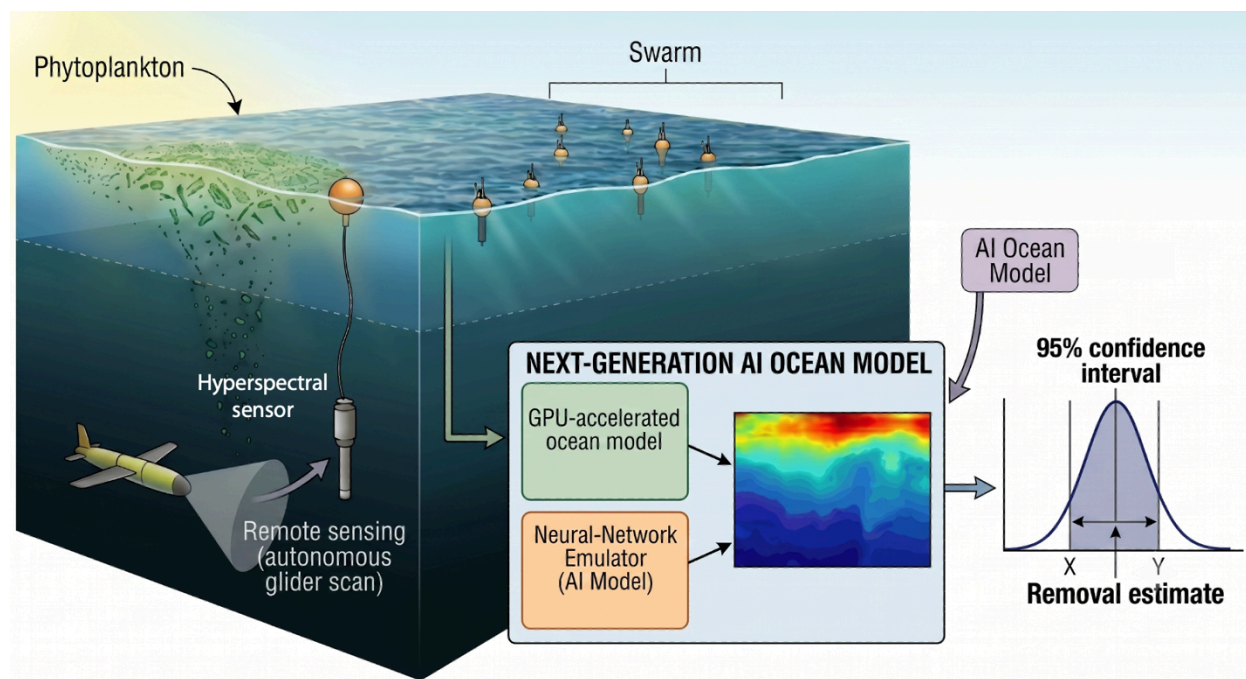


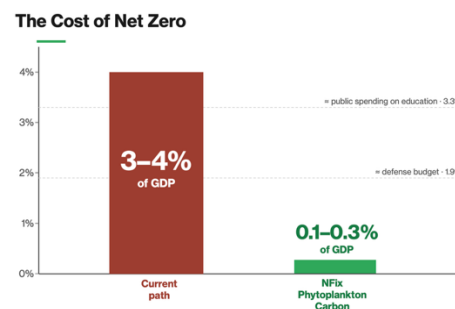
NFix: Creating Value from Phytoplankton Carbon Removal

Phytoplankton can remove carbon from the atmosphere at scale with unmatched efficiency. This program will clear the main bottleneck to commercializing a new Phytoplankton Carbon Removal approach, by developing new sensors and models that produce the kind of data markets can price.



1. An unsustainable path

Japan must cut emissions by 73% by 2040 and down to zero by 2050 by law. The current strategy is economically unviable. In hard-to-abate sectors like steel, chemicals, cement, aviation and shipping, decarbonizing via green hydrogen and green ammonia would cost \$110-\$160 billion per year: 3-4% of GDP — as much as Japan spends on education, twice what it spends on defense.¹ A next generation phytoplankton pathway — NFix Carbon Removal — will enable a transformatively affordable alternative, by building technology that allows markets to price phytoplankton carbon credits.



2. Faster, Wealth-Generating Decarbonization

Microscopic marine photosynthesizers naturally remove from carbon dioxide from the air at scale and store it durably as they sink into the deep. But they could produce much more. Around half of the surface of the

¹ Sector abatement costs from R.T. Shafiee & D.P. Schrag, "Carbon abatement costs of green hydrogen across end-use sectors," *Joule* 8 (2024): 3281–3289 (steel, chemicals, cement); Thunder Said Energy, "Decarbonize shipping: alternative fuel costs" (green ammonia); and ICCT / Carbon Direct sustainable-aviation-fuel cost analyses (aviation). Sector emissions from Japan's national greenhouse-gas inventory and manufacturing statistics; GDP from the World Bank (2024). Author's own calculations: each sector's annual emissions (≈ 246 MtCO₂ in total) multiplied by its green-hydrogen or green-ammonia abatement cost ($\approx \$150$ – $1,300$ /tCO₂) yields $\approx \$110$ – 160 billion per year, or 2.8–3.9% of Japan's $\approx \$4.0$ trillion GDP.

earth is covered by marine deserts that sustain very little life. These waters could sequester carbon at multi-gigaton scale if enough nitrogen were fixed in the upper layer of the ocean to sustain phytoplankton.

Across the earth's warm, subtropical oceans — including Japan's Ogasawara Islands and Minamitorishima— small nutrient additions to the water, designed feed the ocean's natural nitrogen fixers, could lead to large expansions in durable carbon removal. Researchers at Growing Oceans, an organization we work closely with, are developing an approach that works by supplying small volumes of inexpensive minerals to surface waters. This keeps costs low: as little as \$10 per ton of CO₂ durably removed from the atmosphere.

3. Measurement unlocks the pathway

The key challenge for phytoplankton-based carbon dioxide removal is accurate measurement. Phytoplankton locks carbon away when it dies and drifts down through the water column. This happens hundreds of meters under the surface in complete darkness. Reliably estimating precisely how much reaches the deep ocean unlock svalue of this carbon lock.

Today, we measure organic carbon export by suspending a plastic funnel, letting it catch sinking particles, then hauling it back onto a ship and measuring it. Accurate, but unscalable across a large body of water. Other methods fall short: satellites can measuresurface chlorophyll but cannot see beneath the first optical depth. Ocean models are too coarse and loosely tied to real data to measure durable removal accurately.

4. Bounding the uncertainty around carbon flux estimates

Ocean carbon fluxes happen deep below the surface, so they must be estimated rather than measured. To produce a carbon flux estimate robust enough to trade on the GX ETS, the key is to reliably quantify uncertainty. The goal of the program is to create a system that identifies the probability distribution around removal estimates, yielding claims of the form “this intervention durably removed between X and Y tons of CO₂, with 95% certainty.” The narrower the spread between X and Y, the more valuable the resulting carbon credit will be.

To produce carbon flux estimates with verifiably bound uncertainty, the program will pioneer:

New Sensors designed specifically for carbon quantification at scale.

- Remotely detect carbon fluxes using sound, electromagnetics or other physics-based techniques.
- Develop in-situ hyperspectral sensors able to measure chemistry across the water column.
- Design low-cost sensors to be deployed in swarms able to cover an entire patch of ocean.

New Models designed to leverage the expected enormous data streams. A new GPU-accelerated ocean model of phytoplankton carbon flux, designed to generate training data for a neural-network emulator, giving rise to a next-generation AI Ocean Model able to assimilate data from the new sensors to provide carbon flux estimates with measured uncertainty

5. Risk Mitigation

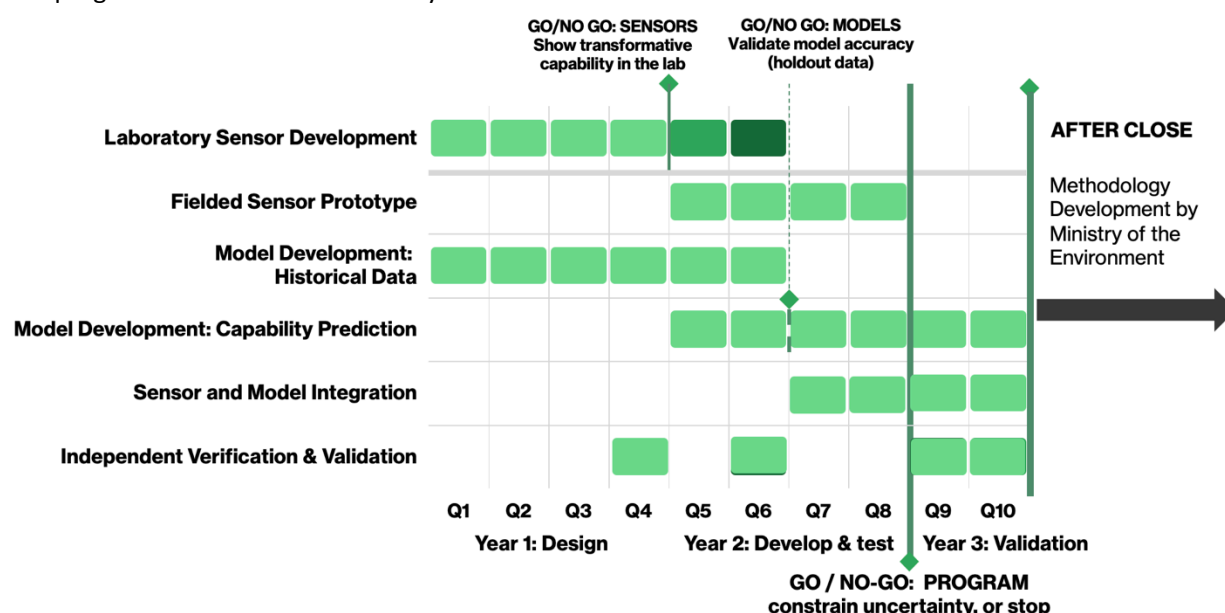
Risk	Probability/Impact	Mitigation Approach
Uncertainty. Keeping model uncertainty within narrow bounds may prove difficult.	High/High	Require an Observation System Simulation Experiment (OSSE) to validate technology accuracy
Durability. Transport to deep ocean is insufficient at test sites.	Medium/High	Consider geographic alternatives through regional modeling of ocean downwelling.

Ecology. Testing leads to negative downstream ecological impacts on key species.	Moderate/Low	Adopt adaptive management practices and dual-use monitoring of carbon flux and ecosystem state.
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The program aims to derisk the world's climate response portfolio. Adverse results would be of scientific interest.

6. Thirty months to a verified measurement protocol

The program will take two and a half years to deliver.



At the end of Q4, we'll reach a Go/No Go gate with regard to sensors: those unable to show transformative capability in the lab will be wound up. At the end of Q6, we'll reach a Go/No Go gate with regard to models: those unable to validate model accuracy with regard to holdout data will end. At the end of Q8 months, we'll reach a Go/No Go gate with regard to the program: if estimate uncertainty cannot be constrained to a usable range, we stop. After 30 months, we'll aim to reach independent verification and validation. The program is ready to be handed off to the Ministry of the Environment, to generate a carbon offset methodology for use in the GX Emissions Trading System.

7. A New Global Industry

The IPCC has stated that "achieving global net zero GHG emissions requires all remaining CO₂ [...] emissions to be counterbalanced by durably stored CO₂ removals." Phytoplankton carbon will dominate the carbon removal industry by the middle of the century because it is more scalable and cost-effective than any other approach. At \$10 per ton of CO₂ durably removed, removing 10 gigatons of CO₂ globally each year would be a new \$100 billion global industry. Because it is the most technically demanding segment of the value chain, monitoring, reporting and verification will be the most lucrative. This program will put Japanese IP, Japanese verification methodologies and Japanese firms at the core of a global strategy to decarbonize not industry, but the atmosphere.